Rezwan Hassan

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Permanent Resident of Canada with Master's in Computer Science.

Professional Summary

Software Developer with a Master's in Computer Science and professional experience in backend development, enterprise simulation systems, applied AI, and machine learning. Experienced with Java, Python, SQL, REST APIs, predictive analytics, LLM integration, RAG, and AI-agent architectures. Currently contributing to military simulation and AI-enabled software, including backend analytics services, explainable ML prototypes, and evaluation and improvement of an enterprise NL-to-SQL database agent.

Highlights of Qualifications and Core Skills

- **Programming & Backend:** Java, Python, SQL, REST APIs, FastAPI, Object-Oriented Programming, backend service development
- **Applied AI & Machine Learning:** Scikit-learn, Random Forest, Decision Trees, predictive analytics, anomaly detection, explainable AI, LLM integration
- **Generative AI & Agents:** Retrieval-Augmented Generation (RAG), NL-to-SQL, prompt/context engineering, AI-agent architectures, A2A protocol, AI evaluation
- **Databases & Data:** MySQL/MariaDB, PostgreSQL, Pandas, schema analysis, data processing, knowledge-base enrichment
- **Development & DevOps:** Docker, Git, GitLab, branching/rebasing/merging, merge requests, CI/CD workflows, Linux/Windows development environments
- **Additional Technologies:** Angular, AWS, Kubernetes, MLflow, Grafana, Prometheus, TensorFlow/Keras, PyTorch, Spark

Work Experience

Full Stack Developer, SimFront, Canada *(Oct, 2024 – Present)*

- Develop and maintain backend services for military simulation and analytics applications using Java, Python, SQL, REST APIs, and relational databases.
- Designed and implemented backend functionality for reporting and analytics features including Combat Power processing, Unit Watch, Holdings Rollup, and operational dashboards, covering data aggregation, filtering, computation, and business logic.
- Improved reliability and consistency across simulation workflows by troubleshooting and resolving backend, processor, integration, and data-flow issues, and contributed to platform features including Direct/Indirect Firing and MIDAS launcher functionality.
- Designed and developed predictive analytics prototypes using Random Forest/Decision Tree models, deterministic decision-path explanations, FastAPI, optional LLM-generated explanations, and Java-to-Python integration.
- Developed an alternative A2A-based AI-agent prototype, separating prediction and LLM-powered explainability into independently callable services for integration with a broader organizational agent architecture.
- Contribute to the evaluation and improvement of an existing RAG-based NL-to-SQL database agent, including schema-ingestion validation, knowledge-base enrichment, SQL-generation guidance, model failure analysis, and automated regression testing.

- Designed and implemented an NL-to-SQL evaluation framework that executes natural-language test suites, validates generated SQL and schema usage, measures category-level performance, and generates structured evidence for SquashTM-based testing. The framework evaluates SQL generation, execution, expected tables/columns, and unsupported requests.

Automation Engineer, MicroFocus, Canada *(May, 2022 – Dec, 2022)*

- Developed and automated testing frameworks using Selenium and JUnit to validate machine learning models and APIs.
- Designed and implemented CI/CD pipelines with Jenkins to streamline the deployment of ML models.
- Collaborated with data scientists to optimize Python-based models, improving prediction accuracy by 15%.
- Managed cloud environments on AWS for scalable deployment of ML pipelines.

Data Scientist Intern, MicroFocus, Canada *(Jan, 2022 – Apr, 2022)*

- Designed and implemented statistical models to detect threat signals in diverse data streams, solving complex business problems.
- Conducted data analysis using SQL, leading to significant insights through pattern extraction and schema aggregation.
- Developed scripts to transform data and label anomalies, crucial for a model demo that impressed the client.
- Deployed statistical anomaly models in MLflow, effectively identifying and resolving system faults.

Mitacs Accelerate Machine Learning Engineer Intern, Ericsson, Canada *(May, 2021 – Dec, 2021)*

- Designed and deployed LSTM and SARIMAX models for time series forecasting and anomaly detection, improving data-driven decision-making.
- Built robust ETL pipelines to process large-scale data from microservices running on Kubernetes.
- Integrated real-time monitoring solutions using Grafana and Prometheus to detect anomalies in mission-critical applications.
- Used AWS SageMaker to deploy machine learning models at scale, reducing deployment time by 40%.

Applied Projects

Stock Price Prediction Using ML and Sentiment Analysis, Carleton University *(Jan 2021 - Apr 2021)*

- Preprocessed Twitter data for sentiment analysis and implemented three LSTM models to predict stock price fluctuations.

Image Orientation Classification and Prediction , Carleton University *(Sept 2020 - Dec 2020)*

- Developed CNN models for predicting angular differences in images, achieving high accuracy and low MSE with custom architectures and MobileNetV2.

Real Estate Property Management System

- Developed a full-stack real estate management app with Spring Boot, React, JWT authentication, and deployed on AWS EC2 using Docker.

Education

Master of Computer Science *(Sept, 2020 – Feb, 2023)*

Carleton University, Ottawa, Canada

GPA : 11.2/12

Thesis Title: Toward an Automated Real-Time Anomaly Detection Engine in Microservice Architectures

Bachelor of Science in Computer Science and Engineering

BRAC University, Dhaka,Bangladesh (*Jan 201 5 - Dec, 2018)*

CGPA: 3.84/4

Thesis Title: Predicting Depression in Bangladeshi University Students

Publications

Predicting Depression in Bangladeshi University Students. IEEE Xplore, 2018. [Read the full paper]

- Used ML algorithms on a custom dataset to predict depression among students and identify key factors.

Bengali Voice-Controlled AI Robot for the Physically Challenged. Best Paper at IHCI 2020. [Read the full paper]

- Developed a robot with image classification, NLP, and object detection capabilities to assist physically challenged users